

Muhammad Safi ur Rehman

AI ENGINEER | AGENT SYSTEMS | APPLIED LLMS

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PROFESSIONAL SUMMARY

AI engineer and software engineering graduate building production multi-agent systems, human-in-the-loop workflows, and secure full-stack AI platforms. Currently developing the Millos.ai construction-estimating agent harness at Zikra Infotech LLC across orchestration, evaluation, containerized execution, multi-tenant security, and cloud delivery. Strong in TypeScript and Python, with hands-on experience across LangChain, Deep Agents, LangGraph, LangSmith, AWS Bedrock, Daytona, Docker, Supabase, Next.js, and Vercel.

CORE EXPERTISE

Agent harness architecture • Multi-agent orchestration • Prompt and skill engineering • Human-in-the-loop systems • LLM evaluation and trace debugging • Multi-tenant backend security • Cloud deployment • Construction estimating AI

PROFESSIONAL EXPERIENCE

Zikra Infotech LLC | Harness Engineer • Millos.ai Team

Aug 2026 – Present

- Engineer the agent harness for an AI-powered millwork estimating platform that turns construction drawings and project context into structured estimates, pricing checks, workbooks, and proposal outputs.
- Design lead-agent and worker-subagent workflows with Deep Agents and LangChain JS, defining delegation patterns, tool and middleware boundaries, file operations, history compaction, and context-management rules.
- Build durable LangGraph workflows with LangSmith hosting and tracing, including checkpoints, authentication checks, run auditing, and pause-and-resume behavior for long-running estimates.
- Implement human-in-the-loop RFI flows and question cards so estimators can resolve drawing ambiguity, supply missing information, and resume an existing run without losing state.
- Evaluate and debug agent behavior from traces and graded runs, distinguishing found, missed, and invented scope to improve prompts, skills, tool use, and estimation reliability.
- Run command execution and the takeoff engine in isolated Daytona sandboxes backed by versioned Docker images; integrate model inference through AWS Bedrock for reproducible agent runs.
- Contribute across the platform: Next.js, React, TypeScript, Vercel, and Univer on the web layer; Supabase and Drizzle ORM for Postgres, Auth, Storage, Realtime, and row-level security; and Python for the estimate ledger and generated deliverables.
- Maintain a pnpm and Turborepo monorepo with Vitest and GitHub Actions, using Claude Code and Codex to accelerate implementation, review, debugging, and test coverage.

ExiTech | Software Developer

Jan 2026 – Mar 2026

- Developed and maintained web-based workflow automation supporting more than five business processes within a small delivery team.
- Automated repetitive work and improved workflows, reducing manual effort by approximately 25% while supporting faster process turnaround and more reliable deployments.

ExiTech | AI Engineer Intern

Oct 2025 – Jan 2026

- Built four end-to-end MVPs with FastAPI, React, OpenAI, and MongoDB, using reusable backend patterns to reduce prototype delivery time by approximately 40%.
- Implemented vector-search retrieval systems, scalable APIs, and interactive frontends for production-oriented LLM use cases.

Quantum Edge LLC | AI Engineer Intern

Jun 2025 – Aug 2025

- Applied predictive modeling, CNNs, RNNs, and GANs across more than five projects; trained and evaluated models on real-world datasets and improved results through architecture selection and hyperparameter tuning.

SELECTED PROJECTS

Clicky Local | *Swift • macOS • Qwen • Codex • Playwright*

- Built a native macOS menu-bar agent with voice and typed input, foreground-app context, and hybrid routing between a local Qwen model and a cloud Codex planner.
- Designed a schema-validated, one-action-at-a-time tool loop with bounded permissions, verification receipts, completion guards, stale-state checks, and safe recovery for file, editor, browser, native UI, and Trello actions.
- Implemented profile-scoped browser automation, approved-folder file operations with read-before-write and undo, a VS Code bridge, Keychain-backed credentials, and prompt-injection-resistant tool boundaries.
- Reached 21 passing Swift tests and live-verified an end-to-end browser flow that navigated, filled, clicked, read back the result, and reported completion only after observation.

AgentForge | *FastAPI • OpenAI • React • MongoDB*

- Built a visual multi-agent workflow platform with more than 50 reusable pipelines and drag-and-drop agent chaining.
- Developed backend orchestration for agent execution and dynamic prompts, plus persistent branching workflows and execution history supporting more than 1,000 runs.

Multi-Tenant RAG Platform | *FastAPI • OpenAI • Vector Search • React*

- Built a platform for more than 20 users to upload documents and deploy embeddable chatbots through script tags or API keys.
- Implemented retrieval, tenant isolation, document management, and secure API access; handled more than 1,000 queries without cross-tenant data exposure.

Cofoundry | *FastAPI • OpenAI • React*

- Built a full-stack AI platform for startup validation, automating market research and competitor analysis and generating structured decision support.
- Created a one-click website generation and deployment workflow to shorten the path from an idea to a working launch page.

TECHNICAL SKILLS

Agent and LLM systems: LangChain JS, Deep Agents, LangGraph, LangSmith, OpenAI models on AWS Bedrock, RAG, prompt engineering, skill engineering, tool design, middleware, context compaction, agent evaluation

Languages: Python, TypeScript, JavaScript, Swift, SQL, C++, Java, C#, Go

Full stack: Next.js, React, FastAPI, Node.js, Univer, REST APIs, Supabase, Postgres, Drizzle ORM, MongoDB, Qdrant

Infrastructure and delivery: Daytona, Docker, Vercel, pnpm, Turborepo, Vitest, GitHub Actions, Git, CI/CD

AI-assisted engineering: Claude Code, Codex, Cursor; AI-assisted implementation, review, debugging, and testing

Applied domain: Millwork takeoff, drawing scales, units, pricing rules, estimate ledgers, workbook and proposal generation

EDUCATION

FAST NUCES | BS Software Engineering

Jun 2026

National University of Computer and Emerging Sciences • Islamabad Campus • CGPA 3.08

Relevant study: Generative AI, Artificial Intelligence, Data Structures and Algorithms, Database Systems, Software Design and Architecture, Cloud Computing

SELECTED TRAINING

LLM Bootcamp, Full Stack Deep Learning 2026 • Prompt Engineering for AI Applications, Alison